Group and solo flight: comparisons

Group time, solo flights and the same flights alone, on open flights

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Abstract

We compare paraglider group time with solo flights and with the solo time of the same flights, on the flights the pilots declare open on the FFVL catalogue, with no stratification. At lags of hours the mean squared displacement of solo flights bends down: the longest lags rest on declared open flights that fold back on themselves, whose takeoff to landing distance is a small part of the path flown. We keep the declared open flights with an opening above 0.1, the lowest threshold that drops those coming back near their takeoff while keeping nearly the same share of lowland and mountain flights (88% and 80%; a threshold of 0.5 would keep 42% and 13%). On these open flights group time progresses straighter than solo flights at every lag, $H_{\mathrm{loc}}(1000\ \mathrm{s})$ of 0.915 against 0.897, the same flights alone move like solo flights, and the bend at long lags is much weaker.

1 Aim

Does a paraglider move differently in a group than alone? This report compares group time with solo flight on open flights, one step at a time, and reads each result together with the number of flights behind it, given in every table and figure.

Section 2 defines the flights, the populations and the measures. Section 3 compares the populations as they are on the flights the pilots declare open, and finds that their mean squared displacement bends down at long lags. Section 4 shows why: many declared open flights fold back on themselves, and the longest lags rest on those. It sets a threshold on how open a flight is, and Section 5 repeats the comparison on the flights that pass it.

2 Flights, populations and measures

Which flights, which stretches of flight, and how are they measured?

Declared open flights. On the FFVL catalogue the pilot declares the type of the flight (Table 1). A distance, free or with up to three turn points, is open; a triangle, a quadrilateral or an out-and-return is closed; a hike and fly is neither. We keep the declared open flights: 66 845 of the 154 598 paraglider flights that depart and are logged every 10 s or finer (87 484 are closed, 269 neither).

label	meaning	class	flights
Dist 3 pts	distance, 3 turn points	open	40 094
Dist 2 pts	distance, 2 turn points	open	25 399
Dist libre	free distance	open	694
Dist 1 pt	distance, 1 turn point	open	658
triangle	triangle	closed	51 181
triangle FAI	FAI triangle	closed	34 979
Quadrilatère	quadrilateral	closed	703
Aller-Retour	out and return	closed	617
Marche et Vol bouclé	hike and fly, closed	closed	3
Marche et Vol FAI	hike and fly, FAI triangle	closed	1
Marche et Vol	hike and fly	neither	269

Table 1: The labels of the FFVL catalogue, their class and their flights (paragliders that depart, logged every 10 s or finer).

Groups. The labels are those of the note *Group flights in free flight* (`methods/group-identification`). Two pilots are in contact when one is on the path of the other within $\delta = 2$ min, $\varepsilon_h = 500$ m horizontally and $\varepsilon_z = 300$ m vertically (headings compared only in glide) for at least $\tau^* = 38.1$ min. The configurations of an instant are the connected components of the contact graph, and a group keeps the same members over a maximal interval. The groups are those of every flight of the archive, whoever the companions.

Populations. After its departure, the first fix 3 km from the takeoff, every instant of a flight is in a group or solo. Three populations:

- S, *solo flights*: flights that never entered a group, the whole of their time after the departure;
- G, *group time*: the stretches in a group of the flights that entered one at least once;
- A, *the same flights alone*: the solo stretches of those same flights, with the same pilots, gliders, loggers and days as G.

Measure. A *piece* is a stretch within one segment of the track, on a 10-s grid of its clock. For a flight f and a population, the time-averaged mean squared displacement pools the origins of all its pieces in that population,

$$\overline{\delta_f^2}(\tau) = \frac{1}{N_f(\tau)} \sum_t \|\mathbf{r}(t+\tau) - \mathbf{r}(t)\|^2, \tag{1}$$

with t and $t+\tau$ in the same piece. The ensemble MSD(τ) is its mean over the flights, one weight per flight. $H_{\mathrm{loc}}(\tau)$ is half the slope of log MSD against log τ over the lags within ± 0.25 decades of τ: 1 for a straight line, $\frac{1}{2}$ for a random walk. Bands are 5–95% of bootstrap draws that resample the dependency blocks of the grouping.

Datasets. A lag τ needs pieces of at least $\tau/0.8$ s, so that the origins of every piece cover the lag in the same share of it. A dataset to the lag T keeps, in each population, its pieces of at least $T/0.8$ s and measures them at every lag up to T: the population does not change along a curve, and each population of a figure has at least 100 flights at T. The pieces of at least 1250 s give the lags up to 1000 s; we add $T \approx 4 \cdot 10^3$ s, the lag of the grid nearest to 4000 s, and the longest T at which each population keeps 100 flights.

3 Group, solo and alone on the declared open flights

How do the three populations compare as they are, with no stratification?

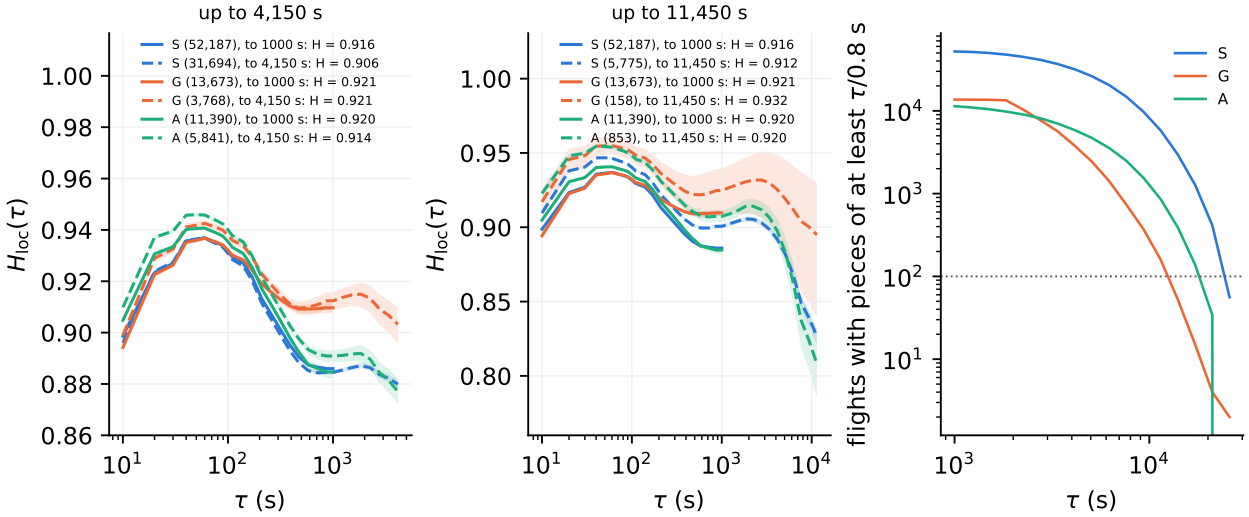


Figure 1: Declared open flights. $H_{\mathrm{loc}}(\tau)$ of S, G and A for the datasets up to 4 150 s (left) and 11 450 s (centre): thin, the pieces of at least 1250 s, up to 1000 s, with their flights in the legend; dashed, the pieces of at least $T/0.8$ s; H in the legend is half the slope of log MSD against log τ over all the lags of the curve; 5–95% bands. Right: flights with pieces of at least $\tau/0.8$ s; dotted, 100 flights. Flights of each curve in Table 2.

Up to 1000 s group time is the straightest: $H_{\mathrm{loc}}(1000\ \mathrm{s})$ is 0.910 in G, against 0.886 in S and 0.885 in A (Figure 1). On the dataset up to 4 150 s the order is the same: at 4 150 s, 0.903 [0.896, 0.910] in G, 0.880 in S and 0.877 in A. On the dataset up to 11 450 s the mean squared displacement of S and A bends down beyond

dataset	S	G	A
lags up to 1000 s	52 187	13 673	11 390
lags up to 4 150 s	31 694	3 768	5 841
lags up to 11 450 s	5 775	158	853

Table 2: Flights of each dataset of Figure 1, declared open flights.

about $5 \cdot 10^3$ s: at 11 450 s H_{loc} is 0.827 in S and 0.808 in A. That of G stays at 0.895 [0.839, 0.930], on 158 flights only.

What we keep. Raw, group time progresses straighter than solo flights and than the same flights alone, and at lags of hours the motion of solo flights and of the same flights alone bends away from a power law.

4 How open the declared open flights are

Why does the mean squared displacement bend down at long lags?

Opening. The *opening* of a flight is the distance from its takeoff to its landing, its first and last fix, over the path flown: 1 for a straight line, near 0 for a flight that comes back to its takeoff.

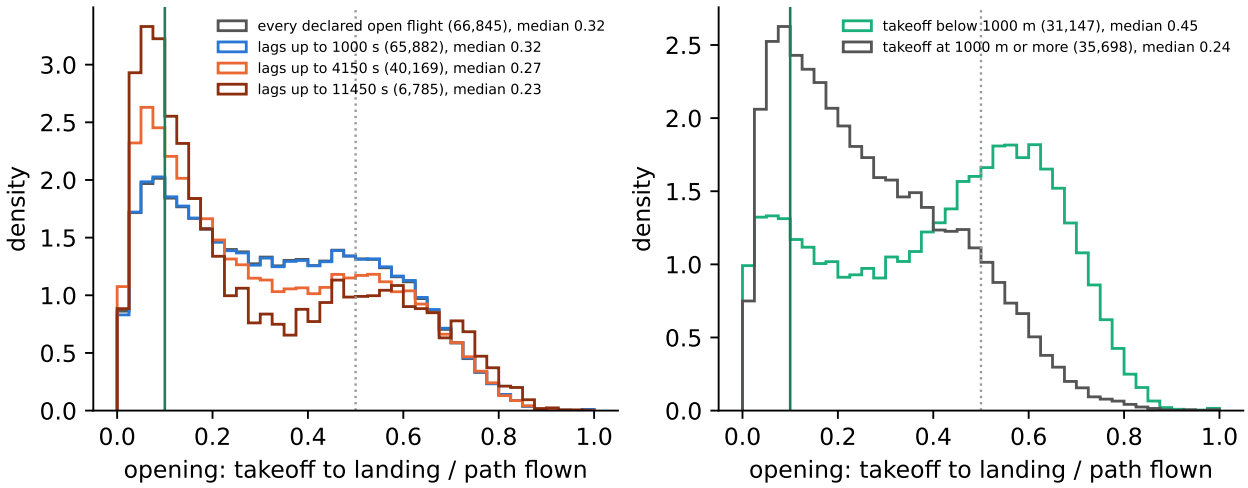


Figure 2: Opening of the declared open flights, takeoff to landing over the path flown. Left: every declared open flight and the flights of the datasets of Figure 1. Right: takeoffs below and above 1 000 m. Solid, the threshold 0.1; dotted, the stricter 0.5 that we do not use.

	flights				median opening			
dataset	all	S	G	A	all	S	G	A
every declared open flight	66 845				0.32			
lags up to 1 000 s	65 882	52 187	13 673	11 390	0.32	0.33	0.28	0.25
lags up to 4 150 s	40 169	31 694	3 768	5 841	0.27	0.28	0.30	0.21
lags up to 11 450 s	6 785	5 775	158	853	0.23	0.23	0.29	0.19

Table 3: Flights and median opening of the datasets of Figure 1, by population; a flight of G and A counts in each population where it has a piece long enough.

The long lags rest on folded flights. The declared open flights are far from straight lines: their median opening is 0.32 (Figure 2, Table 3). The longer the lag, the more folded the flights it rests on: the median opening is 0.32 for the dataset up to 1000 s, 0.27 up to 4 150 s and 0.23 up to 11 450 s, and the share of flights with an opening below 0.1 grows from 16% to 26%. A flight that comes back near its takeoff moves away and then back, and its displacement stops growing as the lag nears its duration: the bend of Section 3 is the shape of the flights. In the longest dataset the flights of G are more open than those of S and A (median 0.29 against 0.23 and 0.19), so part of the straighter progress of G at long lags is their shape.

The threshold. H has a meaning where the mean squared displacement is a power law, a straight line in log–log, and we want it on flights that are open. A threshold on the opening drops the declared open flights that are not. A strict one, 0.5, keeps only straight flights and raises H by construction; it also keeps 42% of the flights that take off below 1 000 m but only 13% of those above, where relief and airspace do not let a flight go straight (Figure 2, right), so it would select the region as well as the shape. We set the threshold at 0.1: the lowest that drops the flights that come back near their takeoff, it keeps nearly the same share of the two, 88% and 80%. It keeps 55 879 flights, 84% of the declared open ones.

What we keep. From here on an *open flight* is a declared open flight with an opening above 0.1.

5 Group, solo and alone on the open flights

How do the three populations compare on the open flights?

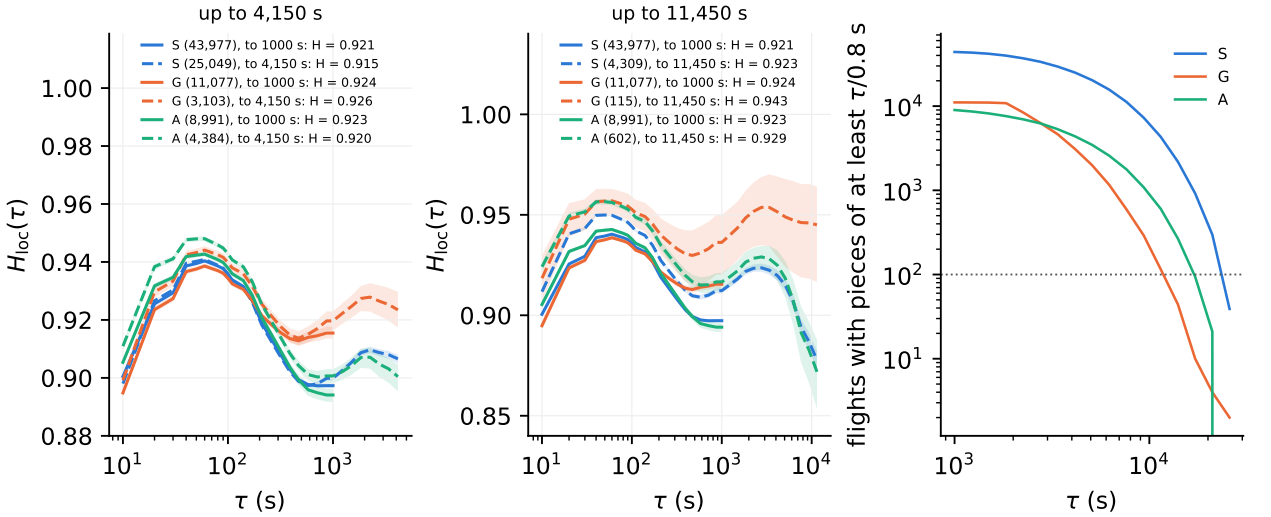


Figure 3: Open flights, opening above 0.1. $H_{loc}(\tau)$ of S, G and A for the datasets up to 4 150 s (left) and 11 450 s (centre): thin, the pieces of at least 1250 s, up to 1000 s, with their flights in the legend; dashed, the pieces of at least $T/0.8$ s; H in the legend is half the slope of log MSD against log τ over all the lags of the curve; 5–95% bands. Right: flights with pieces of at least $\tau/0.8$ s; dotted, 100 flights. Flights of each curve in Table 4.

dataset	S	G	A
lags up to 1000 s	43 977	11 077	8 991
lags up to 4 150 s	25 049	3 103	4 384
lags up to 11 450 s	4 309	115	602

Table 4: Flights of each dataset of Figure 3, open flights.

Group time is the straightest on every dataset (Figure 3): $H_{loc}(1000\text{ s})$ is 0.915 in G, against 0.897 in S and 0.894 in A; at 4 150 s, 0.923 [0.917, 0.930] against 0.907 and 0.900. The same flights alone follow solo flights at every lag. The bend at long lags is much weaker than on the declared open flights: at 11 450 s H_{loc} is 0.878 in S and 0.872 in A, against 0.827 and 0.808 before the threshold. That of G is 0.945 [0.917, 0.964], on 115 flights only.

What we keep. On open flights, raw, group time progresses straighter than solo flights at every lag, while the same flights alone move like solo flights; the threshold weakens the bend at lags of hours.

6 What we keep

- Many flights declared open fold back on themselves, and the longest lags rest on those: the mean squared displacement then bends with the shape of the flights, not with the group.
- An opening above 0.1 drops the declared open flights that come back near their takeoff and keeps nearly the same share of lowland and mountain flights; a stricter threshold would select the region.

- On the open flights, raw, group time (G, 11 077 flights) progresses straighter than solo flights (S, 43 977) up to lags of hours, and the same flights alone (A, 8 991) move like solo flights.

The comparison is raw: group and solo flights differ in route, season, hour and wing, which the next comparisons fix.